Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/06/2014
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Optical Computer Recognition of Stress, Affect and Fatigue in Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
Start Date:	06/01/2013	**End Date:**	05/31/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date change to 5/31/2016 per NSBRI (Ed., 11/18/2014) NOTE: End date change to 11/30/2016 per NSBRI (Ed., 9/30/14) NOTE: End date changed to 5/31/2016 per NSBRI submission on 6/3/2014 (Ed., 6/23/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mollicone, Daniel (Pulsar Informatics, Inc.) Metaxas, Dimitri (Rutgers University) Basner, Mathias (University of Pennsylvania) Goel, Namni (University of Pennsylvania)		
Grant/Contract No.:	NCC 9-58-NBPF03401		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronaut performance during spaceflight can be compromised by stress, negative emotions, and fatigue. This project is focused on a way to detect these behavioral states during long-duration missions through development and validation of an objective, unobtrusive, computational model-based tracker using optical computer recognition (OCR) that reliably identifies facial expressions of negative and positive emotions, and fatigue evident in slow eyelid closures. OCR technology being developed will unobtrusively acquire data on positive, neutral, and negative emotional expressions during spaceflight to help medical officers and astronauts be aware of and manage behavioral health during missions, when communication delays make current approaches to behavioral health less feasible. OCR development and validation is being accomplished via collaboration between Dr. Dinges (Unit for Experimental Psychiatry, University of Pennsylvania Perelman School of Medicine), and Dr. Metaxas (Computational Biomedicine Imaging and Modeling Center, Rutgers University). The project has primary relevance to strategic goals of the NSBRI Neurobehavioral and Psychosocial Factors Team and NASA Behavioral Medicine program. Following OCR algorithm development, laboratory validation, and system score ratings of "high" by 2 former astronauts, NSBRI had the project plan re-scoped from the laboratory to NASA's Human Exploration Research Analog (HERA), to complete a "proof-of-principle study" (letter to PI on 4-29-2013). During year 01 of this re-scope, OCR algorithm technology was further enhanced and work was performed integrating it into NASA's Campaign 1 HERA missions 1 (Feb. 2014) and 2 (Apr. 2014). The PI was notified that based on prior work performed, the second project goal (i.e., detection of performance risks from fatigue via OCR) had been achieved, and the focus of the proof-of-principle study was to be on detection of emotional expressions (email to PI on 4-4-2014) in N=8 HERA crewmembers. In year 01, OCR video data were acquired throughout both of the two 7-day Campaign 1 HERA missions on N=4 crewmembers per mission (N=8 total). Video data were collected approximately 16h per day of each 7-day mission (i.e., 7d x 16h = 112h of video per mission) when crews were not in their sleeping quarters, where there were no cameras allowed. The target sample size (N=8) was studied, and excellent progress was made on all 3 specific aims in year 01. Specific Aim 1, to test the technical feasibility of acquiring facial video data for OCR analyses on N=8 HERA crewmembers, was achieved with 111 hours(h) of video data collected during HERA-1 (HERA-2 video data acquisition is underway through April 30, 2014, and will yield a comparably complete video data set). A total of 89h (80%) of the 111h of video data acquired in HERA-1 was useable for OCR analyses (the remaining 20% was unusable due to crewmember movement out of frame or location with no camera). The high rate of usable video data for OCR analyses was made possible by the Dinges Lab development in the first half of year 01 of a sophisticated system for OCR video data acquisition, transmission and storage, which they installed and their technical personnel managed throughout both HERA-1 and 2 missions in close collaboration with NASA engineers. Specific Aim 2 focused on determining the validity of OCR accuracy for detecting facial expressions of HERA crews by double-blind comparisons with humans using the Facial Expression Coding System (FACES) to score emotional expressions in the same HERA videos. Acquisition and validation of OCR data were accomplished by using OCR video of HERA crewmembers pre-mission to train (i.e., adjust) the OCR algorithm on the facial expressions of each individual; and (2) facilitation of OCR video data capture and storage in HERA-1 and 2 by technical personnel from the Dinges Lab remaining at HERA before and during the missions. Preliminary analyses of HERA-1 results indicated that OCR algorithm analyses of the HERA-1 crewmembers' facial expressions of emotions had a sensitivity of 80% and a specificity of 64% (overall accuracy 71%). Going forward, OCR algorithm adjustments will be undertaken for increases in sensitivity and specificity (to achieve an accuracy =85%), including additional OCR algorithm pre-mission training on crewmembers' faces, and use of other OCR epoch time bases (current results are for 15s scoring epochs). In addition to data on OCR and human scoring of facial expressions of positive and negative emotions in HERA-1 and 2, crewmembers' visual analog ratings of tiredness, fatigue, mental and physical exhaustion (AM/PM), as well as sleep quality, conflicts and social desirability were acquired for comparison with our extensive data on these measures from ISS and Mars 520. HERA-1 crewmembers had very few negative reactions (owing to a relatively benign mission scenario), but what was seen was consistent with what the PI has observed in data he has acquired from ISS and Mars 520 (Basner et al., PLoS One, 2014). Specific Aim 3, to make changes to the OCR system to optimize functionality, was accomplished by algorithm improvements in the Metaxas Lab in year 01. The first software enhancement was to accommodate in-plane rotations of human faces as are likely to occur in microgravity (e.g., face appearing upside down). The second addressed the challenge of facial landmark localization and tracking from a single camera as crewmembers moved about. The resulting OCR algorithm framework devised by the Metaxas Lab can now simultaneously handle face detection, pose-free landmark localization and tracking in real time (see Yu et al., IEEE International Conference on Computer Vision, 2013). In year 02 (2014-2015) we plan on completing OCR data evaluation in HERA-2, continue improvements of the OCR algorithm, and gather additional OCR data for validation in 3 new HERA missions to bring the total sample size to N=20.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study focuses on the ability of an unobtrusive, automated optical computer recognition (OCR) technology to detect psychological stress, emotion, and fatigue in astronauts during space flight. The knowledge gained has the potential to identify an objective, unobtrusive, automated method for the recognition, monitoring, and management of the risks of neurobehavioral dysfunction in affect and alertness in space flight and in many Earth-based safety-sensitive occupations, such as NASA mission controllers, first responders, transportation workers (e.g., truck drivers, train conductors, airline pilots); operators in safety-sensitive industries (e.g., power plant control rooms); and military personnel.
	The following tasks were accomplished in Year 01 of the project. 1. At the request of NSBRI, the project plan was re-scoped from the laboratory to NASA's Human Exploration Research Analog (HERA), to complete a "proof-of-principle study" (letter to PI on 4-29-2013). 2. Extensive work was performed by the Dinges Lab integrating OCR video capture into the HERA facility at JSC, for maximum reliable data acquisition in NASA's Campaign 1 HERA missions 1 (Feb. 2014) and 2 (Apr. 2014). Working in close collaboration with NASA engineers, they installed and managed throughout both HERA missions, a sophisticated system for OCR video data acquisition, transmission, and storage at the HERA facility. They were on-site at HERA to manage all OCR data acquisition before, during, and after each mission (800h). 3. OCR video data was acquired throughout both of the first two 7-day Campaign 1 HERA missions on N=4 crewmembers per mission (N=8 total). Video data were collected approximately 16h per day of each 7-day mission (i.e.,

Task Progress:	7d x 16h = 112h of video per mission) when crews were not in their sleeping quarters, where there were no cameras allowed. A total of 111 hours of video data for OCR analyses were collected during HERA-1 (HERA-2 video data acquisition was underway through April 2014, and has yielded a comparably complete video data set). Initial evaluation of the OCR videos from HERA-1 reveled that a total of 89h (80%) of the 111h of video data acquired was useable for OCR analyses (the remaining 20% was unusable due to crewmember movement out of frame or location with no camera). 4. Videos from HERA-1 were run through the OCR algorithm for emotional expressions at the Metaxas Lab, following training of the algorithm using video of HERA crewmembers pre-mission. Results of the OCR algorithm scoring of video data from HERA-1 were compared (double-blind) to the results of human scorers of the videos, who were trained in the Facial Expression Coding System (FACES). Relative to human scorers, the OCR algorithm had a sensitivity of 80% and a specificity of 64% (overall accuracy 71%). HERA-2 OCR data evaluation will be undertaken in May 2014, and OCR algorithm adjustments will be undertaken to further increase its sensitivity and specificity. 5. Crewmembers' ratings of tiredness, fatigue, mental and physical exhaustion (AM/PM), as well as sleep quality, conflicts, and social desirability were acquired in HERA-1 and 2. Data were 96-100% complete. 6. The OCR algorithm was further enhanced at the Metaxas Lab to (1) accommodate in-plane rotations of human faces as are likely to occur in microgravity, and (2) to simultaneously handle face detection, pose-free landmark localization and tracking in real time.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Basner M, Dinges DF, Mollicone DJ, Savelev I, Ecker AJ, Di Antonio A, Jones CW, Hyder E, Kan K, Morukov BV, Sutton JP. "Psychological and behavioral changes during confinement in a 520-day simulated interplanetary mission to Mars." PLoS One. 2014 Mar 27;9(3):e93298. eCollection 2014. http://dx.doi.org/10.1371/journal.pone.0093298 ; PubMed PMID: 24675720; PubMed Central PMCID: PMC3968121 , Mar-2014
Articles in Peer-reviewed Journals	Yu X, Huang J, Zhang S, Yan W, Metaxas D. "Pose-free facial landmark fitting via optimized part mixtures and cascaded deformable shape model." 2013 IEEE International Conference on Computer Vision (ICCV). 2013:1944-51. (2013 IEEE International Conference on Computer Vision (ICCV), Sydney, Australia, December 1-8, 2013) http://dx.doi.org/10.1109/ICCV.2013.244 , Dec-2013
Awards	Dinges DF. "Overseas Fellow of International Association of Traffic and Safety Sciences, June 2013." Jun-2013